

Animal Nutrition Large Language Modeling Systems (ANLLMS)

Strategic Vision

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Animal agriculture is entering a new era of data-driven decision making. Nutritionists today rely on sophisticated nutrition models, laboratory analyses, precision livestock technologies, and an ever-expanding body of scientific research. Yet these resources remain fragmented across disconnected software platforms, spreadsheets, technical literature, and proprietary databases.

The challenge facing our industry is no longer generating knowledge—it is connecting knowledge.

ANLLMS is a model-agnostic AI orchestration platform designed to unify trusted nutrition models, scientific literature, laboratory analyses, farm management systems, and precision livestock technologies within a secure conversational workspace. Rather than replacing existing software or professional expertise, ANLLMS amplifies both by allowing users to work with the models and data sources they already trust.

Modern nutrition professionals spend valuable time searching technical documentation, reconciling information from multiple software systems, and translating new research into practical recommendations. Meanwhile, feed commonly represents **40–50% of livestock production costs**, meaning even modest improvements in decision quality can create meaningful economic returns. ANLLMS shortens the path from evidence to action while protecting proprietary information through private AI environments.

As DairyLac expands into the United States, it has an opportunity to demonstrate leadership not only through innovative nutritional products, but also by helping shape the future of intelligent technical service. ANLLMS is designed to connect—not replace—the trusted tools and expertise that already drive animal agriculture.

Roadmap to a Minimum Viable Platform (RMVP)

Phase 0 — Founding Industry Partnerships

Advisory partners, pilot scope, governance

Phase 1 — AI Knowledge Workspace

Secure AI over DairyLac technical literature with citations

Phase 2 — Integrated Dairy Pilot

Read-only integration of farm data and nutrition models

Phase 3 — Connected Farm Intelligence

APIs for sensors, herd software, and multi-species support

Phase 4 — Industry Platform

Open, model-agnostic ecosystem for partners

Guiding Principles

• Model-agnostic • Evidence-based • Secure by design • Human-centered • Interoperable

“The future of animal nutrition will not be defined by a single model or a single company. It will be built by connecting trusted science, trusted technology, and trusted expertise through a common intelligence platform.”